Class: BE **Subject: Applied Data Science Lab** **Sem-VIII**

Experiment No 1

Course Outcome:CO1 **Blooms Level: L4**

Aim: To perform descriptive statistical analysis on a real-world time-based dataset using statistical computing and data analysis tools in order to evaluate central tendency, variability, distribution characteristics, and risk-causing outliers for effective decision-making under uncertainty.

Abstract:
In real-world event planning, uncertainty in human arrival behavior can lead to financial loss due to resource overutilization and scheduling inefficiencies. Given a dataset representing the arrival times of individuals to a booked venue, the problem is to statistically determine the typical arrival behavior, quantify the degree of variability and inconsistency, identify the directional imbalance in arrivals, detect the presence of extreme abnormal behaviors, and isolate the individual(s) responsible for maximum financial risk. The study further aims to evaluate whether planning decisions should rely on mean-based or median-based estimation in the presence of outliers, using computational analysis through spreadsheet and Python-based analysis tools.

The analysis should address the following key questions:

- What are the mean and median arrival times of the group, and what do they indicate about the typical arrival behavior and the influence of extreme values?
- What do the standard deviation and coefficient of variation reveal about the consistency, dispersion, and reliability of arrival behavior?
- Does the distribution of arrival times exhibit skewness and high kurtosis, and what do these measures indicate about directional imbalance and extreme behaviors?
- Which individual(s) are identified as statistically abnormal using Z-score analysis, and how do they contribute to overall financial and scheduling risk?
- In the presence of extreme arrival values, should planning decisions be based on the mean or the median, and why?

Theory:

Descriptive Statistics:

Descriptive statistics can be defined as the measures that summarize a given data, and these measures can be broken down further

1. Measure of central tendency
2. Measure of spread/dispersion
3. Measure of symmetry/shape

Measure of Central Tendency

Measure of central tendency is used to describe the middle/centre value of the data.
Mean, Median, Mode are measures of central tendency.

1. Mean

- Mean is the average value of the dataset.

- Mean is calculated by adding all values in the dataset divided by the number of values in the dataset.
- We can calculate the mean for only numerical variables.

2. Median
- The Median is the middle number in the dataset.
- Median is the best measure when we have outliers.

3. Mode
The mode is used to find the common number in the dataset.

Measure of spread

- The measure of spread/dispersion is used to describe how data is spread. It also describes the **variability** of the dataset.
- **Standard Deviation, Variance, Range, IQR,** are used to describe the measure of spread/dispersion
- The measure of spread can be shown in graphs like **boxplot**.

1.Variance
- Variance is used to describe how far each number in the dataset is from the mean.
- Formula to calculate population variance

$$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$$

2.Standard Deviation
- Standard Deviation is the measure of the spread of data from the mean.
- Standard deviation is the square root of variance.
- More the standard deviation, more the spread.

3.Range
- The range is the difference between the largest number and the smallest number.
- Larger the range, the more the dispersion.

4. Interquartile range (IQR)
- Quartiles describe the spread of data by breaking into quarters. The median exactly divides the data into two parts.
- **Q1(Lower quartile)** is the middle value in the first half of the sorted dataset.
- **Q2**– is the median value
- **Q3 (Upper quartile)** is the middle value in the second half of the sorted dataset
- The interquartile range is the difference between the 75th percentile(Q3) and the 25th percentile(Q1).
- 50% of data fall within this range.

Boxplot is used to describe how the data is distributed in the dataset. This graph represents five-point summary (minimum, maximum, median, lower quartile, and upper quartile) and is used to identify **outliers**.

- whiskers—denote the spread of data
- box—represents the IQR- 50% of data lies within this range.

Measure of shape

1.Skewness

Skewness, which is the measure of the symmetry, or lack of it, for a real-valued random variable about its mean. The skewness value can be positive, negative, or undefined. In a perfectly symmetrical distribution, the mean, the median, and the mode will all have the same value.

2.*Kurtosis*

Kurtosis provides a measurement about the extremities (i.e. tails) of the distribution of data, and therefore provides an indication of the presence of outliers. Kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution. That is, data sets with high kurtosis tend to have heavy tails, or outliers. Data sets with low kurtosis tend to have light tails, or lack of outliers.

Program:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
df=pd.read_excel(r'/content/drive/MyDrive/ADS LAB/Party_Data.xlsx')
```

```
df
```

	Person	Time
0	1	60
1	2	67
2	3	58
3	4	63
4	5	56
5	6	70
6	7	120
7	8	65
8	9	67
9	10	61
10	11	56
11	12	67
12	13	1
13	14	67
14	15	66

Next steps: Generate code with df New interactive sheet

```
df["Time"].describe()
```

	Time
count	15.000000
mean	62.933333
std	22.920567
min	1.000000
25%	59.000000
50%	65.000000
75%	67.000000
max	120.000000

dtype: float64

Measures of Central Tendency

Double-click (or enter) to edit

```
m=df["Time"].mean()
print(m)
```

```
62.93333333333333
```

```
med=df["Time"].median()
print(med)
```

```
65.0
```

```
mod=df["Time"].mode()
print(mod)
```

```
0    67
Name: Time, dtype: int64
```

```
harmean=stats.hmean(df["Time"])
print(harmean)
```

```
12.34826593971098
```

Measures of Dispersion

```
m1=df["Time"].max()
```

```
m2=df["Time"].min()
```

```
range=m1-m2
print(range)
```

```
119
```

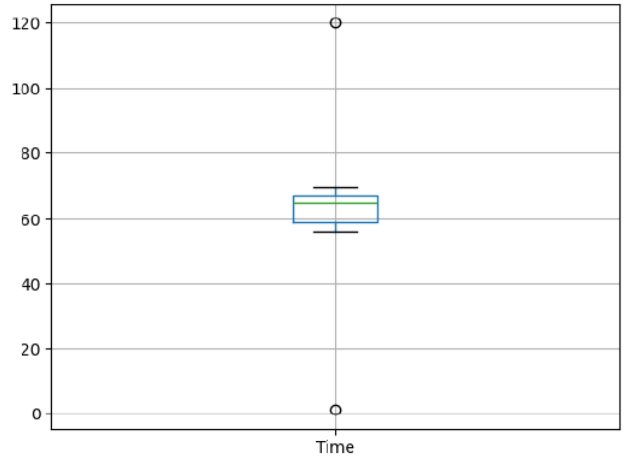
```
Q3=df["Time"].quantile(0.75)
Q1=df["Time"].quantile(0.25)
```

```
IQR=Q3-Q1
print(IQR)
```

```
8.0
```

```
df.boxplot(["Time"])
```

```
<Axes: >
```




```
s=df["Time"].std()
print(s)
```

```
22.92056676769536
```

```
s1=df["Time"].var()
print(s1)
```

```
525.3523809523809
```

```
cv=s/m
print(cv)
```

```
0.36420392109685423
```

```
dq=harmean/m
```

```
print(dq)
```

```
0.19621185285557705
```

```
Measures of Shape
```

Double-click (or enter) to edit

```
skew=df["Time"].skew()
print(skew)
```

```
-0.35506704824442753
```

```
kurtosis=df["Time"].kurtosis()
print(kurtosis)
```

```
6.315098227335365
```

```
zscore=stats.zscore(df["Time"])
```

```
print(zscore)
```

```
[-0.13247004  0.18365165 -0.22279053  0.00301068 -0.31311101  0.31913238
  2.57714447  0.09333117  0.18365165 -0.0873098  -0.31311101  0.18365165
 -2.79692431  0.18365165  0.13849141]
```

Conclusion: The analysis of the group's arrival times shows that the typical arrival is around 65 minutes, as indicated by the median, while the mean is slightly lower at 62.93 minutes due to the influence of an extremely early arrival. The high standard deviation of 22.92 minutes and coefficient of variation of 36.42% reveal considerable variability, indicating that the group is inconsistent and unreliable for precise scheduling. The distribution exhibits a slight negative skew (−0.36) and very high kurtosis (≈6.32), reflecting a cluster of arrivals around 60–70 minutes but with increased likelihood of extreme early or late arrivals. Z-score analysis identifies Friend 7 (120 minutes) and Friend 13 (1 minute) as statistically abnormal, representing potential financial risks, with extreme lateness being the costlier issue. Overall, planning decisions should rely on the median rather than the mean, as it more accurately reflects typical behavior and is robust to the influence of outliers, ensuring better resource allocation and scheduling.

Exercise 1: Explore the descriptive statistics on the given dataset.
Dataset: In this experiment, a fictitious data of Body Mass Index (BMI) containing 10 observations and 5 variables is used. The dataset contains Height, Weight, Age, BMI, and Gender columns.
Abstract:

- What is the typical BMI level of the study group, and how much variability exists in body mass across individuals falling within the clinical risk ranges of 25–27 (pre-diabetic), 28–30 (hypertensive–cholesterol), and ≥30 (obesity–cardiac)?
- Does the distribution of BMI exhibit asymmetry and heavy-tailed behavior due to the presence of individuals in the extreme obesity category (BMI ≥ 30), indicating concentrated cardiac risk within the group?
- Are any individuals statistically abnormal based on BMI Z-scores, particularly in the obesity range (BMI ≥ 30), and do these individuals emerge as true medical risk outliers capable of distorting overall group health assessment?
- Which individual BMI values act as dominant health-risk contributors—analogous to high-cost late arrivals in event planning—by disproportionately inflating the group's average BMI and overall disease-risk perception?
- If preventive health planning must be based on a single robust statistical measure, should the group intervention strategy rely on mean BMI or median BMI to minimize misclassification caused by extreme obesity values (BMI ≥ 30)?

Exercise 2: Exercise: Descriptive Statistical Analysis of the Wine Dataset
Dataset:
A wine quality dataset containing physicochemical properties such as fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, alcohol, sulphates, density, pH, and a sensory quality score (0–10).

Abstract:

- What is the typical alcohol content and quality score of the wine samples, and how much variability exists in these attributes across wines categorized as low quality (≤4), medium quality (5–6), and high quality (≥7)?
- Does the distribution of alcohol content or volatile acidity exhibit skewness or heavy-tailed behaviour, particularly due to the presence of wines with exceptionally high alcohol levels or acidity, indicating concentrated quality-enhancing or quality-degrading effects?
- Are there any wines that can be classified as statistical outliers based on Z-scores of alcohol content, sulphates, or volatile acidity, and do these extreme values correspond to unusually high-

or low-quality ratings?

- Does the presence of wines with very high alcohol content cause a noticeable difference between the mean and median alcohol values, indicating the influence of extreme observations?
- How large is the spread (standard deviation and interquartile range) of alcohol content, and does it suggest that most wines cluster around a typical alcohol level or are widely dispersed?